Logistic Regression (LR)

Logistic Regression is a supervised machine learning algorithm used for classification problems. It predicts the probability that a given input belongs to a particular class. Unlike linear regression, which predicts continuous values, logistic regression predicts categorical outcomes such as Yes/No, True/False, or Approved/Rejected.

The algorithm uses the sigmoid function to convert predicted values into probabilities between 0 and 1. Based on this probability, the model assigns the input to a specific class. Logistic regression is commonly used in applications such as medical diagnosis, spam detection, and loan approval prediction.

Dataset Description

The dataset used for this Logistic Regression model is a loan approval dataset. It contains information about loan applicants and their financial characteristics. The dataset includes attributes such as income, loan amount, credit history, and other relevant details that help determine whether a loan should be approved.

Each row represents a loan applicant, and each column represents a specific attribute related to that applicant. The dataset contains independent variables (input features) that describe the applicant's details and a dependent variable (target class) that indicates whether the loan is approved or rejected. This dataset is suitable for logistic regression because the problem involves binary classification.

STEPS IN PYTHON IMPLEMENTATION

Step 1:

```python
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
```

In this step, the required Python libraries are imported. NumPy is used for numerical computations, Pandas is used for handling and processing the dataset, and Matplotlib is used for visualization. These libraries are essential for data analysis and machine learning implementation.

Step 2:

```python
dataset = pd.read_csv('loan_approval.csv')
```

```python
dataset
```

In this step, the dataset is loaded using the Pandas read_csv() function. The dataset is stored in the variable dataset, which allows the data to be accessed and analyzed in the following steps.The loaded dataset is displayed.

Step 3:

```python
X = dataset.iloc[:, :-1].values
y = dataset.iloc[:, -1].values
```

In this step, the dataset is divided into independent variables (X) and the dependent variable (y). The independent variables contain the input features used for prediction, while the dependent variable represents the target class that the model will learn to predict.

Step 4:

```python
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=0)
```

In this step, the dataset is divided into training and testing sets. The training data is used to train the logistic regression model, while the testing data is used to evaluate its performance. In this case, 80% of the data is used for training and 20% for testing.

Step 5:

```python
from sklearn.preprocessing import StandardScaler

sc = StandardScaler()

X_train = sc.fit_transform(X_train)
X_test = sc.transform(X_test)
```

Feature scaling is applied to standardize the range of input features. This ensures that all features contribute equally to the model and prevents features with larger values from dominating the prediction process.

Step 6:

```
from sklearn.linear_model import LogisticRegression

classifier = LogisticRegression()

classifier.fit(X_train, y_train)
```

```
LogisticRegression
LogisticRegression()
```

In this step, the Logistic Regression model is created using the Scikit-learn library. The fit() function is used to train the model using the training dataset so that it can learn the relationship between the input features and the target class.

Step 7:

```
y_pred = classifier.predict(X_test)

print("Predicted values:", y_pred)
print("Actual values:", y_test)
```

```
Predicted values: [1 0 0 1]
Actual values: [1 1 0 1]
```

In this step, the trained logistic regression model is used to predict the class labels for the testing dataset. The model predicts whether the loan application will be approved or rejected.

Step 8:

```
from sklearn.metrics import accuracy_score

accuracy = accuracy_score(y_test, y_pred)

print("Accuracy:", accuracy)
```

```
Accuracy: 0.75
```

This step calculates the accuracy of the logistic regression model using the accuracy_score() function. Accuracy measures the percentage of correct predictions made by the model.